

Neural Networks

Structure, Components & How They Learn

AIML-501 | Portfolio Artifact



Exploring the building blocks of artificial intelligence

What is a Neural Network?

Definition

A neural network is a computational system modeled loosely on the structure of the human brain. It consists of interconnected layers of nodes (neurons) that process information, detect patterns, and learn from data.

Neural networks are the foundation of modern AI — powering image recognition, language models, recommendation systems, and more.

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Layers in a
deep network

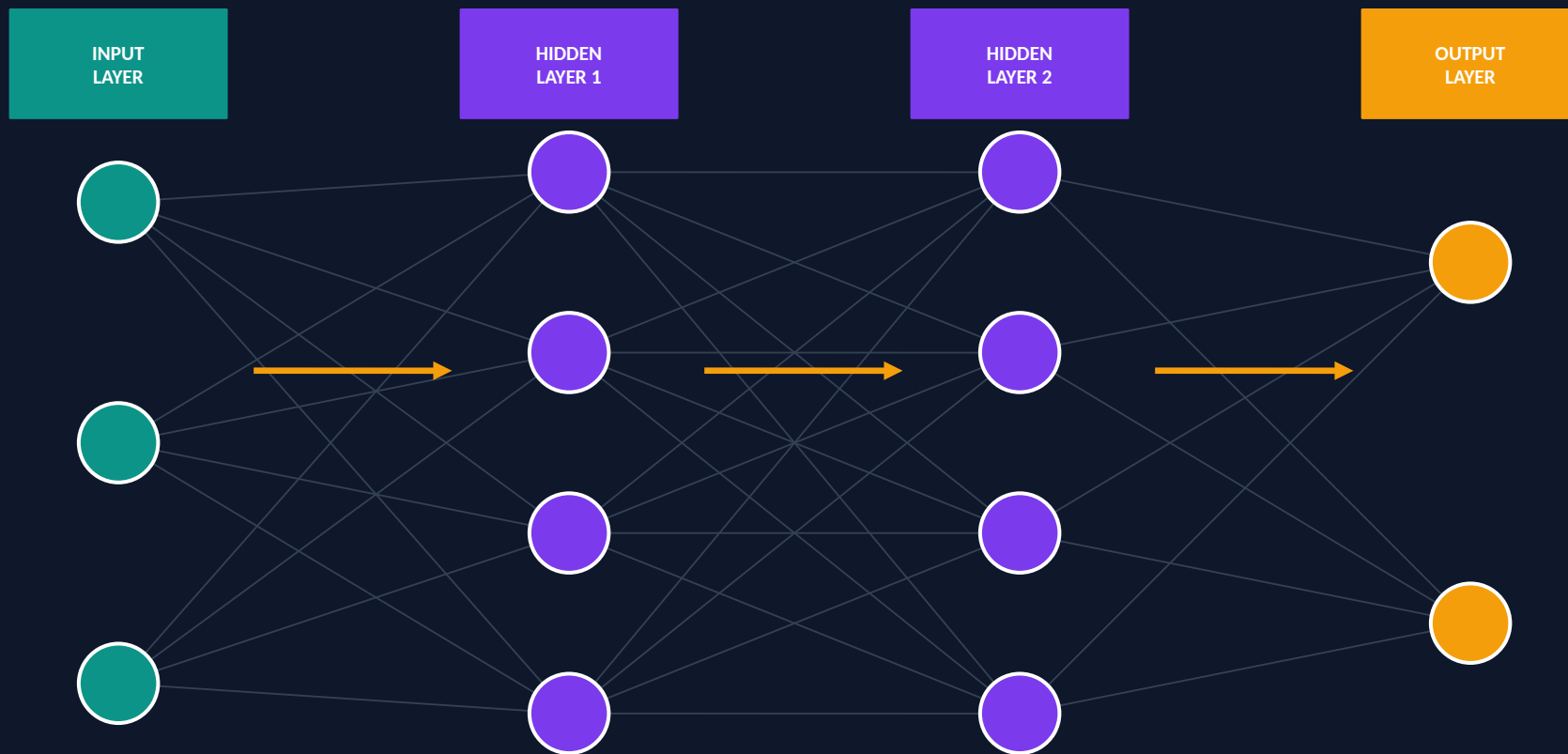
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Patterns it
can learn

1940s

Origin of the
perceptron

Neural Network Architecture — Data Flow



Data flows left → right through each layer, transforming inputs into predictions

Components: Layers & Neurons



Layers

Neural networks are organized into layers — groups of neurons that process data at the same stage.

- Input Layer: Receives raw data (pixels, words, numbers)
- Hidden Layers: Extract progressively abstract features
- Output Layer: Produces the final prediction or classification

Deeper networks (more hidden layers) can learn more complex patterns, but also require more data and compute.



Neurons

A neuron is the basic computational unit of a neural network. Each neuron:

- Receives inputs from previous neurons
- Multiplies each input by a weight
- Sums all weighted inputs plus a bias term
- Passes the result through an activation function

Inspired by biological neurons, but far simpler — a neuron is essentially a weighted sum followed by a nonlinear transformation.

Components: Weights & Activation Functions



Weights

Weights are numerical parameters that control the strength of connections between neurons. They determine how much influence one neuron's output has on the next.

- Initially set randomly
- Updated during training via backpropagation
- A high weight amplifies a signal; a near-zero weight suppresses it

Learning in a neural network is fundamentally the process of finding the right weights. A bias term (an additional adjustable constant) is also added to each neuron to shift the activation threshold.



Activation Functions

Activation functions introduce non-linearity into the network, allowing it to learn complex, curved decision boundaries that linear models cannot represent.

Common types:

- ReLU (Rectified Linear Unit): $\max(0, x)$ — most widely used in hidden layers
- Sigmoid: squashes output to 0–1 — used for binary classification output
- Softmax: converts outputs to probabilities summing to 1 — used for multi-class output
- Tanh: squashes output to -1 to 1

Components: Loss Functions & Optimization



Loss Functions

A loss function (also called a cost function) measures how far the network's predictions are from the true answers. It produces a single number — the loss — that summarizes total error across all training examples.

Common types:

- Mean Squared Error (MSE): for regression tasks — penalizes large errors heavily
- Cross-Entropy Loss: for classification — measures the difference between predicted and actual probability distributions

Training goal: minimize the loss.



Optimization Algorithms

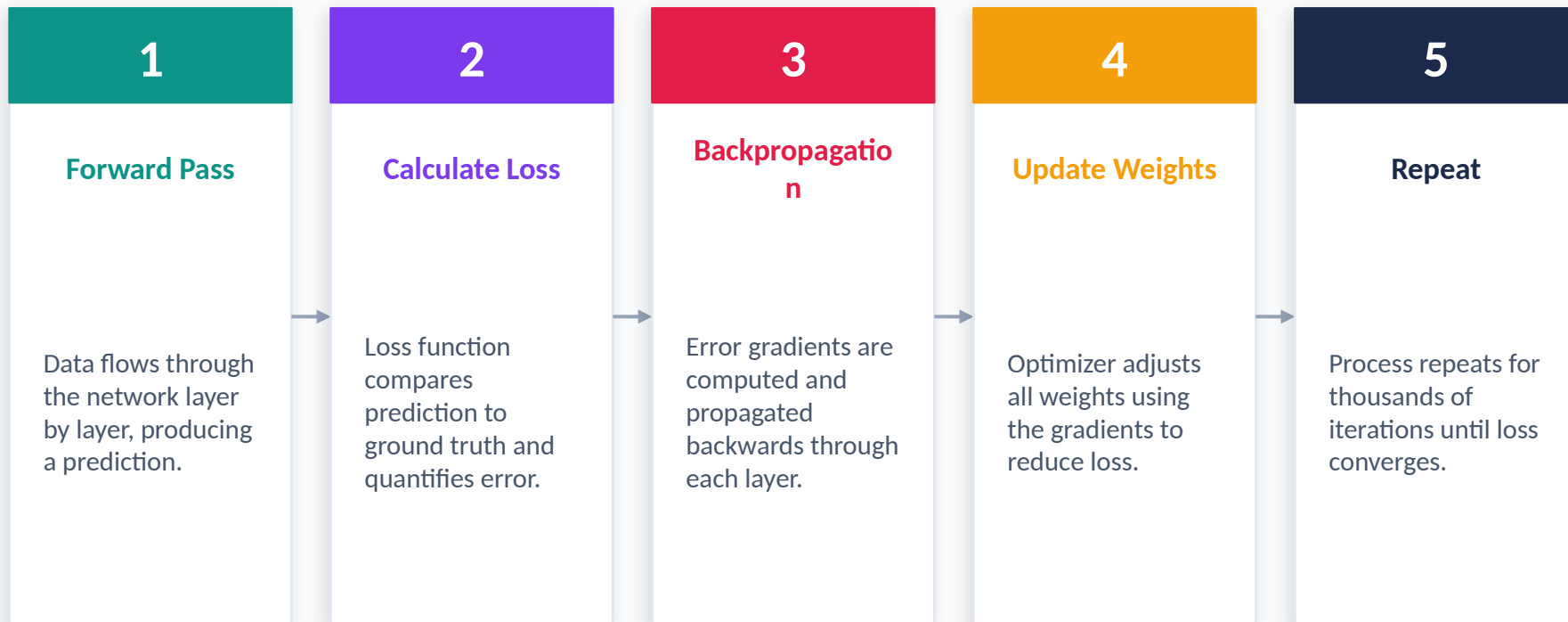
Optimization algorithms adjust the network's weights to reduce the loss function. The most foundational method is Gradient Descent — moving weights in the direction that reduces the loss most steeply.

Key variants:

- SGD (Stochastic Gradient Descent): updates weights using one sample at a time — fast but noisy
- Mini-batch GD: uses small batches — balances speed and stability
- Adam: adapts the learning rate per parameter — the most widely used optimizer today

Learning rate controls the step size; too large causes instability, too small causes slow convergence.

How a Neural Network Learns — The Training Loop



This cycle — forward pass → loss → backprop → update — is repeated until the model's predictions are accurate.

Summary & Key Insights

6 Core Components

Layers:	Organize neurons into input, hidden, and output stages
Neurons:	Compute weighted sums and pass results through activation functions
Weights:	Learnable parameters adjusted during training to minimize error
Activation Functions:	Introduce non-linearity (ReLU, Sigmoid, Softmax)
Loss Functions:	Quantify prediction error (MSE, Cross-Entropy)
Optimization:	Adjust weights to reduce loss (SGD, Adam)

Key Insight

Neural networks do not follow explicit rules. They discover patterns by iteratively adjusting millions of weights until predictions match the training data.

The power of deep learning comes from stacking many layers — each layer building more abstract representations of the input, from raw pixels to edges to shapes to objects.

Every AI system you interact with — from search engines to language models — is built on these six foundational components.